

## PAPER 1 / QUESTION 109

25 April 2024 / TZ1 (file label) / HL Paper 1 / original question 4 / 2224-6701

---

**109.** An object with mass  $m$  falls through the air with a terminal speed  $v$  for a short time  $t$ .

What is the work done on the air by the falling object during this time?

A.  $mgvt$

B.  $mgt$

C.  $\frac{mgv}{t}$

D.  $\frac{mv}{t}$